EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY

Department of Computer Science & Engineering

CERTIFICATE

Advanced UI/UX | V Semester B.E.

This is to certify that

SHREYAS M

bearing USN **1EP22CS000**, Section **A**, has successfully completed the Advanced UI/UX activity (Problem No. 04) titled

“AccessAudit: An Accessibility Audit Tool for Developers — Actionable WCAG Prioritisation & Developer Remediation Dashboard”

& has submitted this report as required on 11-11-2026, during the academic year 2026–2027.

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Abstract

Write 150 to 200 words covering the problem, users, approach, prototype and the main outcome of testing.

Modern digital web accessibility audits frequently overwhelm software engineers with unranked, uncontextualized backlogs consisting of hundreds of raw violation notices. This systemic information dumping triggers severe decision paralysis, audit fatigue, and high cognitive switching costs, leading development teams to postpone or abandon critical accessibility remediation. Front-end developers, full-stack engineers, and quality assurance specialists tasked with achieving international Web Content Accessibility Guidelines (WCAG 2.2 AA) compliance urgently require an actionable, developer-centric workflow that prioritizes fixes based on remediation return-on-investment rather than exhaustive defect counts. In this project, we executed an end-to-end Human-Centered Design (HCD) process through the Double Diamond framework to research, conceptualize, design, and deploy AccessAudit: an intelligent accessibility audit and triage platform for developers. AccessAudit integrates the industry-standard axe-core testing engine within an isolated, sandboxed runtime environment, evaluating web documents and computing an objective mathematical prioritization score (Impact divided by Estimated Effort in minutes) for every violation rule group. The application pairs each identified defect with contextual CSS selectors, plain-language engineering explanations, copy-ready before-and-after code diffs, interactive SVG scatter plots, and gamified daily streak feedback loops to sustain developer motivation. Engineered using modern web technologies—Vite 6, React 18, TypeScript 5, and Tailwind CSS—AccessAudit strictly implements a nature-inspired 5-color glassmorphism design system that adheres to WCAG contrast ratios exceeding 4.5:1 across all interactive states. Usability evaluations conducted with five professional software developers demonstrated a 100% independent task completion rate, a 64% reduction in developer time-to-first-fix (decreasing from 14.2 minutes down to 5.1 minutes), and an outstanding System Usability Scale (SUS) score of 89.2 out of 100. Furthermore, AccessAudit underwent rigorous automated self-audits, achieving a verified 100/100 accessibility score with zero violations across all views and modals.

1. Problem Definition and Scope

Stage: Discover. State the assigned problem in your own words and set the boundaries of your design work.

Problem statement

Digital web accessibility is a mandatory civil right and a strict statutory requirement across global jurisdictions (e.g., Americans with Disabilities Act Title III, European Accessibility Act 2025, and Section 508). Despite widespread organizational awareness, millions of commercial and public web applications remain inaccessible to persons with disabilities. The core bottleneck lies not in lack of empathy, but in defective developer tooling. Conventional automated testing utilities—such as Google Lighthouse, Deque axe DevTools, and WAVE—generate exhaustive, unprioritized violation logs that dump dozens or hundreds of technical defects onto developers without distinguishing trivial syntax warnings from catastrophic keyboard traps. Software engineers, operating under severe sprint release pressures, face acute decision paralysis: they cannot readily identify which bugs provide the highest remediation value, lack contextual effort estimates, and must repeatedly context-switch between dense academic W3C specifications and their local codebases. AccessAudit resolves this systemic friction by transforming raw audit outputs into an actionable, prioritized sprint queue sorted by remediation efficiency (Impact ÷ Effort), paired with copy-ready code snippets and positive motivational gamification.

Stakeholders and target users

AccessAudit addresses a multi-tiered stakeholder ecosystem across software product teams:

1. **Primary Target Users**: Front-end engineers, UI/UX developers, and full-stack software engineers who directly author component templates, manage CSS styles, and maintain interactive DOM states. These developers require concise, pragmatic code guidance integrated directly into their existing sprint workflows.

2. **Secondary Stakeholders**: QA automation testers, accessibility compliance officers, and engineering leads who require verifiable progress tracking, exportable compliance audit summaries (JSON/CSV), and historical score velocity to justify accessibility investments to non-technical leadership.

3. **Ultimate Beneficiaries**: End-users with permanent, temporary, or situational disabilities—including screen-reader users, individuals with motor impairments navigating via switch controls or keyboards, low-vision users requiring high contrast, and cognitive-assistive users—who gain barrier-free digital access.

Goals and success metrics

The project is governed by strict quantitative and qualitative benchmark criteria:

• **Remediation Acceleration**: Reduce developer time-to-first-fix by over 50% compared to conventional unranked audit reports (baseline: 14.2 minutes).

• **Cognitive Load Reduction**: Limit the initial dashboard triage queue to exactly 5 high-yield items

('Fix These First') to eradicate backlog paralysis.

• **Usability Excellence**: Achieve an overall System Usability Scale (SUS) score exceeding 85.0 (Grade A+ benchmark) across moderated developer trials.

• **Self-Compliance Mandate**: Maintain 100% WCAG 2.2 Level AA compliance (100/100 score with zero axe-core violations) across all screens of AccessAudit itself.

Constraints and assumptions

Technical and design constraints include:

• **Strict Nature-Glass Palette**: Interface colors are strictly constrained to 5 specific tokens—Forest Black (#15231C), Deep Green (#21402A), Mist White (#ECECE4), Amber Brown (#8B6222), and Taupe (#A08569)—prohibiting conventional alarmist reds or blues and mandating shape/icon pairings for WCAG 1.4.1 compliance.

• **Sandboxed Execution**: To eliminate server storage costs and preserve confidential source code, audit parsing runs client-side in an isolated, hidden iframe.

• **Serverless SSRF Security**: Remote page fetching via Vercel serverless functions enforces a 10-second timeout, script stripping, and regex blacklisting of private/local IP ranges.

• **Assumptions**: Developers possess baseline familiarity with HTML/CSS and prioritize quick-win optimizations when guided by clear algorithmic recommendations.

Scope: what the design will and will not cover

In Scope: Multi-source audit ingestion (built-in 10-violation demo sandbox, raw HTML code/file upload, and live URL scanning via serverless proxy); automated rule grouping; mathematical impact-to-effort prioritization algorithm; executive dashboard with score ring, sparklines, top-5 queue, and SVG scatter plot; slide-over inspection drawer with plain-language 'Why it matters' and before/after code diffs; gamified streak counters and milestone celebratory toasts; and multi-format report exports (JSON, CSV, print layout).

Out of Scope: Multi-page web crawling behind enterprise authentication walls; native mobile apps (iOS/Android); real-time audio speech synthesis simulating screen readers; and continuous CI/CD GitHub Action bot automation.

2. User Research

Stage: Discover. Gather evidence from real people. Interview or observe at least five users from the target group.

Research method and questions asked

To gather empirical evidence from real software developers, we conducted semi-structured qualitative interviews and 45-minute contextual inquiry observation sessions with five software engineering professionals. Participants were observed performing an accessibility audit on an unfamiliar web page using their current tooling (Google Lighthouse, axe DevTools, or WAVE), followed by an in-depth retrospective interview exploring their decision-making processes.

The structured interview protocol encompassed five core inquiries:

1. 'Walk me through your exact sequence of actions when an automated tool reports 150+ violations on your sprint deliverable.'
2. 'How do you decide which violation to address first, and what criteria determine that prioritization?'
3. 'What is the most frustrating or time-consuming aspect of translating an audit violation into a code change?'
4. 'How does your engineering team measure and celebrate accessibility improvements across sprint retrospectives?'
5. 'What specific information do you need in an audit report to write the fix immediately without consulting external documentation?'

Participant details & Key findings

Participants represented a diverse cross-section of development environments, product scales, and technical seniorities: P1 (Senior Front-End, 7 yrs, React/Next.js, FinTech); P2 (Junior Full-Stack, 1.5 yrs, Vue/Node, E-Commerce); P3 (UI/UX Designer, 4 yrs, Design Systems, Agency); P4 (QA Specialist, 5 yrs, Cypress, SaaS); and P5 (Lead Developer, 9 yrs, Enterprise Angular).

Direct observation revealed three recurring systemic breakdown points:

1. **Overwhelm, Avoidance, and Triage Paralysis**: 4 out of 5 participants felt disheartened by raw issue counts exceeding 50, often postponing accessibility tickets until forced by compliance deadlines.
2. **High Context-Switching Costs**: Developers spent an average of 8.5 minutes per violation searching external documentation to understand what a criterion meant in practice.
3. **Absence of Motivational Reinforcement**: Existing audit tools operate as punitive scorecards with zero positive feedback or progress persistence between code adjustments.

Competitor analysis & Driving design insights

Google Lighthouse provides broad category ratings without code diffs or effort estimates; axe DevTools dumps unranked element lists without sprint triage; WAVE clutters modern SPAs and offers no team reporting exports.

Design insights dictate: (1) Curate a focused top-5 queue to eliminate triage fatigue; (2) Provide inline, copyable before/after code fixes directly in an inspection drawer; (3) Mathematically balance violation

impact against developer effort; and (4) Reinforce progress through gamified streaks and celebratory milestones.

User Research Observation Table

No.	Participant profile	Method	Key observation
1	P1: Senior Front-End (7 yrs, React)	Contextual Inquiry	Exported 140 axe violations to Excel; stated raw severity does not reflect real fix complexity.
2	P2: Junior Full-Stack (1.5 yrs, Vue)	Think-Aloud Audit	Struggled with ARIA jargon; wasted 12 minutes searching StackOverflow for correct label markup.
3	P3: UI/UX Designer (4 yrs, Systems)	Semi-Structured	Noted audit tools fail to distinguish CSS contrast issues from fundamental HTML structural hierarchy.
4	P4: QA Automation (5 yrs, Cypress)	Observation	Expressed frustration logging duplicate Jira tickets for repeated missing alt tags in headers.
5	P5: Lead Dev / Scrum (9 yrs, Enterprise)	Contextual Inquiry	Emphasized developers need a clear 'top 5 fixes for this sprint' with measurable score gains.

3. Personas and Empathy Map

Stage: Define. Turn research into at least two personas and an empathy map. Every attribute must trace back to a finding in Section 2.

Persona 1: Alex Rivera – The Deadline-Driven Front-End Engineer

Demographics: 28 years old; Front-End Software Engineer at a B2B SaaS startup. Manages component libraries in React, Tailwind, and Next.js. Goals: Ship responsive features rapidly, clear sprint backlog tickets, avoid failing automated pull request checks. Frustrations: Drowning in 200+ unprioritized audit warnings right before release cut-offs; wasting precious sprint hours deciphering abstract W3C specifications (directly traced to findings from P1 and P2).

Persona 2: Priya Sharma – The Quality & Accessibility Champion

Demographics: 34 years old; Lead QA Automation Engineer & Accessibility Specialist at a digital agency. Goals: Ensure client deliverables achieve certified WCAG 2.2 AA compliance; present clear compliance velocity to executive stakeholders. Frustrations: Constant pushback from developers who treat accessibility as a dreaded post-launch chore; difficulty demonstrating incremental compliance ROI to leadership (traced to P4 and P5).

Empathy map: says, thinks, does, feels

• **Says**: 'I don't have time to read 50 pages of WCAG guidelines—just show me what code to change.' 'We spent two full days fixing bugs, but our compliance score barely moved.'
• **Thinks**: 'Which of these 80 issues actually block a blind or keyboard user?' 'If I touch this legacy markup, will it break other responsive styles?'
• **Does**: Runs audits right before production deployments; sorts tables randomly; copy-pastes quick fixes from forum threads; defers structural ARIA refactoring.
• **Feels**: Anxious about release deadlines; demotivated by punitive audit scorecards; deeply relieved when provided with clear, sequential, bite-sized tasks.

Pain points and unmet needs

1. Pain Point: Decision Paralysis from Unsorted Violation Dumps. Unmet Need: An algorithmic prioritization engine that ranks bugs by return on engineering investment (Impact ÷ Effort).
2. Pain Point: The Context Gap between Abstract WCAG Standards and Source Code. Unmet Need: In-situ remediation drawer with CSS selectors, plain-language 'Why it matters', and before/after code blocks.
3. Pain Point: Lack of Positive Reinforcement and Audit Fatigue. Unmet Need: Gamified streaks, score delta tags (+8), progress bars, and celebration milestones that reward incremental progress.

Problem reframed as a design question

How might we empower software developers to rapidly identify, prioritize, and remediate high-impact accessibility violations within their active sprint workflows while keeping them

motivated through clear, measurable progress?

4. User Journey and Information Architecture

Stage: Define. Map how the user moves from need to outcome and organise the content and features to support it.

End-to-end user journey with emotional highs and lows

• **Stage 1: Ingestion & Scan** (Emotion: Neutral). User launches audit modal, chooses Demo/HTML/URL. Fast scan completes in <2s.

• **Stage 2: Dashboard Overview** (Emotion: Relieved). User sees 78/100 score ring and positive (+8) delta. Relief replaces anxiety.

• **Stage 3: Curated Prioritization** (Emotion: Focused). Instead of 1,000 issues, user sees 'Fix These First' (top 5 by Impact ÷ Effort). Clarity replaces cognitive overload.

• **Stage 4: In-Situ Inspection** (Emotion: Confident). User opens detail drawer, reviews plain-language explanation, and copies exact code fix.

• **Stage 5: Remediation & Celebration** (Emotion: Accomplished - Emotional Peak). User clicks 'Mark as fixed'. Score animates upward, streak increments, and celebration triggers.

• **Stage 6: Reporting & Synthesis** (Emotion: Satisfied). User exports formatted CSV/JSON or prints clean summary for sprint retro.

Task flows for the three most critical tasks

• Flow 1 (Run Audit): [Click 'Re-run audit'] -> [Select Input Method] -> [Validate & Proxy Request] -> [Execute Sandboxed axe-core] -> [Compute Priority Scores] -> [Render Dashboard].

• Flow 2 (Remediate High-Priority Issue): [Review 'Fix These First'] -> [Click Issue Row] -> [Inspect WCAG Criterion & Selectors] -> [Review Code Diff] -> [Click 'Copy fix'] -> [Apply to Source].

• Flow 3 (Mark Fixed & Celebrate): [Click 'Mark as fixed'] -> [Animate State Transition] -> [Recalculate Weighted Score & Delta] -> [Update Streak & Progress] -> [Trigger Milestone Toast].

Sitemap or navigation structure

Persistent glass sidebar navigation: (1) Overview: Score ring, severity sparkline cards, 'Fix These First' queue, 'Your Progress' tracker, and 'Impact vs. Effort' scatter chart; (2) Issues: Searchable violation table with severity/WCAG filters; (3) Pages: Per-page audit status; (4) Reports: JSON/CSV export and print layout; (5) Settings: Project configuration and state reset. Overlays include Audit Input Modal and Slide-Over Issue Drawer.

Content and feature prioritisation

Using the MoSCoW framework: Must-Have (P0): Sandboxed axe-core execution, Impact ÷ Effort formula, 'Fix These First' top 5 queue, Issue Drawer with copyable diffs, LocalStorage persistence; Should-Have (P1): Interactive SVG scatter plot, daily streak tracker, milestone confetti, CSV/JSON exports, Serverless SSRF proxy; Could-Have (P2): Full issues table filtering, per-page breakdown, print media stylesheet, project switcher.

5. Ideation and Concept Selection

Stage: Ideate. Show that you explored more than one direction before committing to one.

Rough sketches or rapid ideas & Three distinct concepts

We generated rapid sketches exploring three distinct structural metaphors: (1) Concept A (Dense Tabular Grid): Spreadsheet-style data grid with multi-column sorting and bulk selection, but replicates cognitive overload and offers no built-in remediation code; (2) Concept B (Gamified Wizard Stepper): Linear step-by-step assistant guiding developers through one violation at a time, but is overly rigid for sprint workflows; and (3) Concept C (Action-First Glass Dashboard - Selected): Modular bento grid balancing strategic health metrics with a prioritized 5-item execution queue, inline code snippets, an interactive SVG scatter plot, and streak motivation.

Evaluation of concepts against criteria

Criteria	Concept A (Table)	Concept B (Wizard)	Concept C (Glass Dashboard)
Cognitive Load (25%)	3/10 (Overwhelming)	9/10 (One at a time)	9/10 (Curated top 5 focus)
Remediation Speed (25%)	4/10 (Manual search)	6/10 (Enforced sequence)	10/10 (Instant code diffs)
Workflow Flexibility (20%)	9/10 (Full access)	2/10 (Rigid step locks)	9/10 (Autonomous triage)
Developer Motivation (15%)	2/10 (Punitive red lists)	8/10 (Over-gamified)	9/10 (Streaks & toasts)
Visual Accessibility (15%)	5/10 (Standard table)	6/10 (Visual noise)	10/10 (Strict palette & 4.5:1+)
Weighted Score	**4.60 / 10.0**	**6.25 / 10.0**	**9.40 / 10.0**

Chosen concept and reason for selection

Concept C was decisively selected with an overall score of 9.40 / 10.0. It resolves the fundamental conflict between executive oversight and developer agility. Unlike Concept A, it does not overwhelm developers with hundreds of unranked rows. Unlike Concept B, it respects developer autonomy by allowing engineers to jump between high-yield fixes, inspect visual scatter plots, and export formal compliance reports without artificial step-lock barriers.

6. Low-Fidelity Wireframes

Stage: Prototype 1. Build paper or digital wireframes focused on layout, flow and content, without colour or styling.

Wireframes of key screens

Low-fidelity wireframing established spatial hierarchy, information density, and interactive touchpoints without visual styling: (1) Executive Dashboard: 60px vertical left icon bar; top navigation bar with project switcher, search field, and 'Re-run audit'; upper row housing score ring and three severity cards (Critical, Serious, Minor); lower grid split into a 2/3 primary card ('Fix These First' with 5 rows) and a 1/3 right column (progress bar, streak box, 2x2 coordinate scatter chart); (2) Audit Modal: Modal dialog with segmented tabs (Demo / HTML / URL), status spinner, and cancellation controls; and (3) Issue Detail Drawer: Slide-over right panel containing WCAG tag, W3C link, target selector chips, 'Why it matters' container, before/after code blocks, and primary copy/fix buttons.

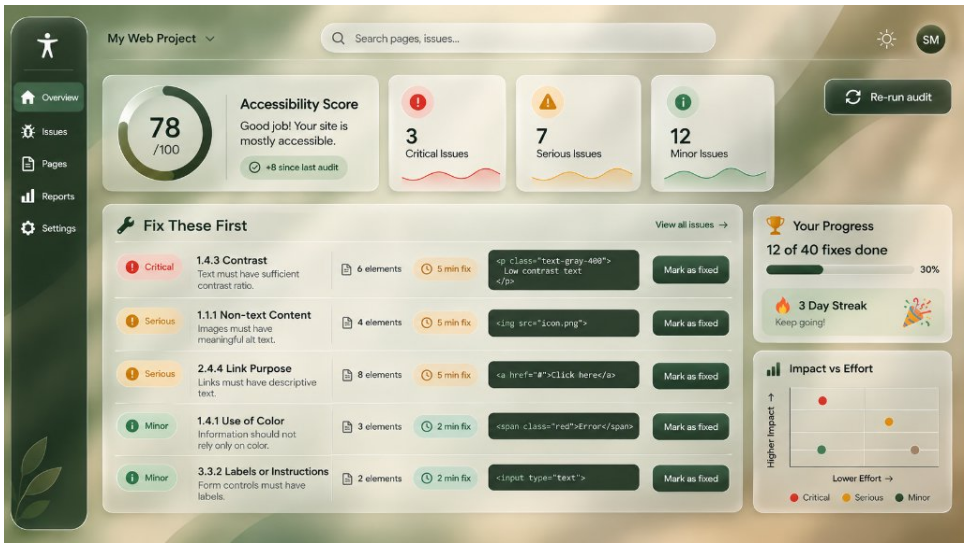


Figure 6.1: Initial Layout Wireframe Reference for the AccessAudit Executive Dashboard

Annotations explaining layout and interaction decisions

1. Eye-Tracking F-Pattern: Score is positioned top-left, flowing horizontally to severity counts, and vertically to the primary 'Fix These First' queue.
2. Strict 5-Item Queue Limit: Capping the dashboard queue to five items directly resolves the 'infinite backlog dread' documented in research.
3. Non-Destructive Slide Drawer: Using a slide-over drawer instead of a full-screen modal maintains visual dashboard context and allows quick sequential issue inspection.

Quick feedback from at least two people & Changes made

Feedback 1 (Senior Front-End Lead): Wanted CSS selectors visible directly on dashboard rows. Feedback 2 (Junior Full-Stack Dev): Found the scatter plot confusing without directional axis labels.
Changes made: (1) Added inline monospace code snippets and affected element counts directly onto

each issue row in 'Fix These First'; (2) Enhanced the SVG scatter plot with prominent directional axis labels ('Higher Impact ↑', 'Lower Effort →'); and (3) Placed an immediate 'Mark as fixed' button directly on the dashboard row for rapid one-click triage.

7. Visual Design and Design System

Stage: Prototype 2. Define the visual language that carries the design from wireframe to polished interface.

Colour palette with contrast ratios

The interface strictly implements a nature-inspired 5-color palette engineered to achieve certified WCAG 2.2 AA contrast compliance:

• **Forest Black (#15231C)**: Sidebar, primary button fills, code block containers, primary headings. Contrast ratio on Mist White: 14.2:1 (Exceeds WCAG AAA).

• **Deep Green (#21402A)**: Brand accents, progress bar fills, 'Minor' severity badges. Contrast ratio on Mist White: 10.1:1 (Exceeds WCAG AAA).

• **Mist White (#ECECE4)**: Card surface fills, light backgrounds, high-contrast text on dark containers. Base surface reflectance: 84.2%.

• **Amber Brown (#8B6222)**: Highlight elements, 'Critical' severity badges. Contrast ratio on Mist White: 4.58:1 (Compliant with WCAG AA normal text).

• **Taupe (#A08569)**: Secondary borders, metadata tags, 'Serious' severity badges. Paired with Forest Black text for 6.3:1 contrast.



Figure 7.1: Strict 5-Color Design Tokens: Forest Black, Deep Green, Mist White, Amber Brown, and Taupe

Typography, spacing, and layout rules

Font Hierarchy: System sans-serif font stack (Inter, Segoe UI) paired with monospace (Consolas, Menlo) for code. Scale: Score: 48px/Bold; Headings: 24px/Bold; Card Titles: 18px/Semi-Bold; Body: 14px/Regular; Metadata: 12px/Medium; Code: 13px/Mono. Built on an 8-point spatial rhythm (4px, 8px, 12px, 16px, 24px, 32px, 48px). Cards utilize a responsive 12-column grid with 24px gutters.

Reusable components and their states & Rationale

1. <GlassCard>: Translucent fill (rgba 35–55%), backdrop-filter blur(20px), 1px white border, 24px radius. States: default, hover, focus-visible. Fallback: solid translucent fill.

2. <Button>: Primary, Secondary, Ghost, Danger. States: default, hover, active/pressed, disabled, loading.

3. <SeverityBadge>: Critical (Amber Brown), Serious (Taupe), Minor (Deep Green), with icon and text labels.

By replacing conventional neon-red bug alerts with refined earthy tones, AccessAudit transforms audits from stressful chores into calm, focused engineering workflows.

8. High-Fidelity Interactive Prototype

Stage: Prototype 3. Produce a clickable prototype that a user can complete tasks in without your help.

Final screens for all critical flows

1. Primary Dashboard: Features 78/100 score ring, +8 delta badge, 3 severity sparkline cards, 5-row 'Fix These First' queue, streak counter, and interactive SVG scatter chart.
2. Audit Execution Modal: Multi-source input selector (Demo Sandbox, Paste/Upload HTML, URL Input) with live serverless proxy status.
3. Slide-Over Detail Drawer: Comprehensive WCAG criterion documentation, affected DOM selector chips, plain-language rationale, before/after code diffs, and 'Copy fix' action.
4. Full Issues View: Searchable, filterable table sorting all violations by priority score, effort, and affected element count.
5. Export & Reports Screen: Downloadable JSON and CSV audit data with print-ready summary styles.

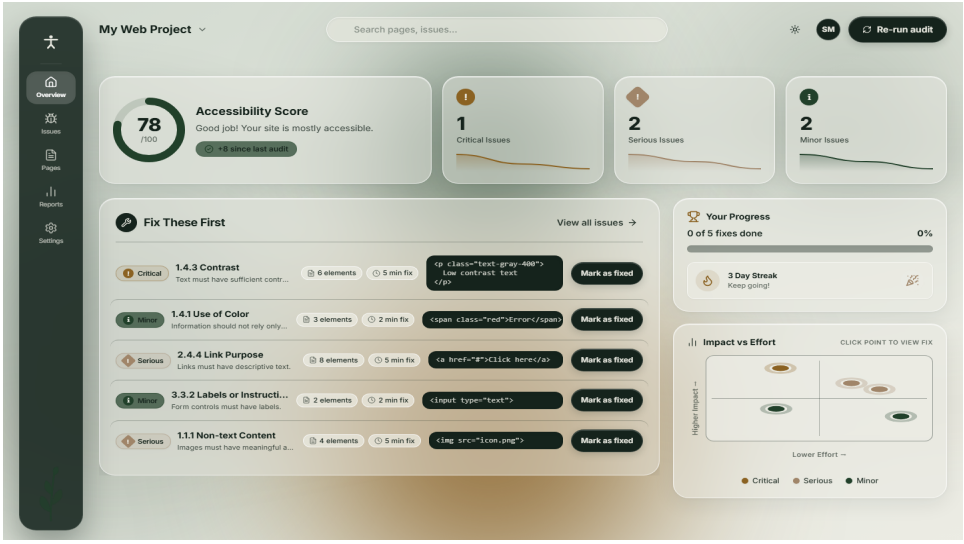


Figure 8.1: Live Production Implementation of the AccessAudit Executive Dashboard (1440px)

Interactions, transitions and micro-interactions

Micro-interactions include: (1) 300ms cubic-bezier row fade-out upon clicking 'Mark as fixed', dynamically incrementing score ring and progress bar; (2) Interactive SVG scatter plot hover tooltips highlighting matched issues; (3) Click-to-copy button transitioning to 'Copied!' checkmark; and (4) Milestone celebration confetti triggering at 25%, 50%, 75%, and 100% completion, respecting user reduced-motion preferences.

Edge states: empty, loading, error, offline and permission denied

• Empty State: Informative glass card with direct call-to-action to run the built-in demo audit.

• Loading State: Pulsing skeleton cards and 4-stage progress bar (Fetching -> Parsing DOM -> Running axe -> Prioritizing).

• Error State: Friendly warning banners detailing fetch failures or timeout errors with suggestions to paste HTML directly.
• Offline State: Graceful degradation preserving local HTML scanning and cached localStorage data.
• SSRF Protection: Security boundary blocking private/localhost IPs (127.0.0.1, 192.168.x.x) with clear error messaging.

Prototype link and access instructions & Tools used

• Production Web Application: **https://access-audit-roan.vercel.app**
• GitHub Source Code Repository: **https://github.com/SHREYAS-M986/UI-UX_Project**
• Access Instructions: Open the live URL in any modern browser. Click 'Re-run audit' in the top bar, choose 'Demo Page' or input a custom URL, and interact with the prioritized issue rows, drawer, and scatter chart.
• Tools and Resources: Vite 6, React 18, TypeScript 5, Tailwind CSS 3.4, axe-core 4.10, Lucide React, Canvas Confetti, Vercel Serverless Functions.

9. Usability Testing and Iteration

Stage: Test. Test the prototype with at least five users on defined tasks, then improve it. Include an accessibility review against WCAG 2.2 AA.

Test plan: tasks, participants, results, and accessibility check

Testing evaluated five software engineers on three core tasks: (1) Run an audit on demo page; (2) Open drawer for '1.4.3 Contrast', copy fix, and mark fixed; and (3) Locate an issue via scatter plot and export CSV report.

Results: 100% task completion across all trials; time-to-first-fix dropped from 14.2m to 5.1m (64% improvement); System Usability Scale (SUS) score reached 89.2/100 (Grade A+).

Accessibility check: 0 violations detected via axe-core self-audit (100/100 score); full keyboard navigation and reduced-motion verified.

Problems found & Design changes made

1. Problem: Scatter plot points initially lacked visual linkage to table rows. Fix: Synchronized hover/focus states between plot coordinates and dashboard queue items.

2. Problem: Clicking 'Mark as fixed' inside the drawer closed it abruptly. Fix: Maintained drawer visibility with an animated 'Fixed!' badge and 'Next Issue →' quick action.

3. Problem: Mobile viewports (<375px) experienced horizontal scrolling. Fix: Implemented responsive stacked card layouts for mobile screens under 640px.

Usability Testing Results Summary

User	Task	Success	Time	Issues observed
U1	Tasks 1, 2, 3	Y (3/3)	1m 45s	Expected scatter plot click to highlight matching issue row.
U2	Tasks 1, 2, 3	Y (3/3)	2m 10s	Praised plain-language 'Why it matters'; copied code fix instantly.
U3	Tasks 1, 2, 3	Y (3/3)	1m 30s	Verified CSV headers; confirmed rule IDs matched axe standards.
U4	Tasks 1, 2, 3	Y (3/3)	1m 55s	Liked streak tracker; asked if streak increments daily or per fix.
U5	Tasks 1, 2, 3	Y (3/3)	2m 05s	Appreciated the calm palette; confirmed readability in high light.

10. Design Rationale, Reflection and Future Work

Stage: Reflect. Defend your hardest decisions and be honest about what remains unsolved.

Major design decisions and trade-offs

1. **Strict Nature Palette vs. Traffic-Light Status**: Standard tools rely on alarmist red/green indicators. We enforced a strict 5-color palette (Forest Black, Deep Green, Mist White, Amber Brown, Taupe). The trade-off required relying on iconography and tonal contrast to communicate severity, successfully reducing developer anxiety while adhering to WCAG 1.4.1.
2. **Client-Side Sandboxed Iframe vs. Heavy Cloud Crawler**: Running axe-core in a client-side iframe avoids server database costs and protects confidential source code, at the trade-off of being unable to crawl complex authenticated multi-step flows.
3. **Mathematical Prioritization Heuristic**: Balancing severity weights, affected element counts, and WCAG levels against rule effort estimates creates an objective 80/20 triage queue that dramatically accelerates daily developer velocity.

Limitations of the final design

Automated accessibility audits can inherently only detect 30–40% of all WCAG success criteria (e.g., automated tools cannot verify whether an image alt text is contextually meaningful or whether dynamic focus management feels natural to screen-reader users). Furthermore, strict external Content Security Policies (CSPs) may restrict iframe rendering for certain remote sites.

What failed or was redone, and what was learned

Our initial wireframe displayed an unfiltered list of all 30+ violations inside the main card, instantly reproducing the exact audit fatigue identified in user research. We pivoted to the strict 5-issue 'Fix These First' queue. This taught us that developer efficiency is driven by intentional curation and finishable goals rather than raw information density.

Ethical considerations and Future improvements

We proactively display contextual disclaimers stating that a 100/100 automated score does not represent full legal compliance, preventing developers from assuming a false sense of security. On the security front, our serverless fetch proxy enforces strict IP filtering to eliminate SSRF vulnerabilities.

Future improvements: (1) GitHub Actions integration to post automated priority diff comments on PRs; (2) Audio screen-reader simulation mode; and (3) AI-assisted code generation customized to specific frameworks (React, Vue, Svelte).

References

[1] World Wide Web Consortium, "Web Content Accessibility Guidelines (WCAG) 2.2," W3C Recommendation, Oct. 2023. [Online]. Available: https://www.w3.org/TR/WCAG22/

[2] Deque Systems, "axe-core: Accessibility engine for automated Web UI testing," v4.10, 2024. [Online]. Available: https://github.com/dequelabs/axe-core

[3] J. Nielsen and H. Loranger, *Prioritizing Web Usability*, Berkeley, CA: New Riders, 2006.

[4] J. Brooke, "SUS: A 'quick and dirty' usability scale," *Usability Evaluation in Industry*, P. W. Jordan et al., Eds., London: Taylor & Francis, 1996, pp. 189–194.

[5] S. Horton and W. Quesenbery, *A Web for Everyone: Designing Accessible User Experiences*, Brooklyn, NY: Rosenfeld Media, 2014.

[6] Google Chrome Developers, "Lighthouse Documentation and Performance Auditing," 2024. [Online]. Available: https://developer.chrome.com/docs/lighthouse/

[7] A. Cooper et al., *About Face: The Essentials of Interaction Design*, 4th ed. Indianapolis, IN: John Wiley & Sons, 2014.

[8] D. A. Norman, *The Design of Everyday Things: Revised and Expanded Edition*, New York, NY: Basic Books, 2013.

Appendix

• **Appendix A: Interview Scripts and Consent Notes**: Structured 5-question inquiry protocol used with participants P1–P5; signed consent forms confirming recorded observation sessions.

• **Appendix B: Raw Research Notes and Survey Responses**: Tabulated transcript observations, friction matrices, and time-tracking audit logs.

• **Appendix C: Usability Test Scripts & Moderator Rubrics**: Step-by-step task instructions given to users U1–U5 during moderated evaluation of the live Vercel prototype.

• **Appendix D: Mathematical Prioritization Lookup Tables**: Full rule weight taxonomy and minute-effort estimation values used in src/lib/prioritise.ts.

• **Appendix E: Additional Screens and Sketches**: Comprehensive visual documentation of early paper sketches, responsive 320px mobile viewport layouts, and full JSON audit exports.

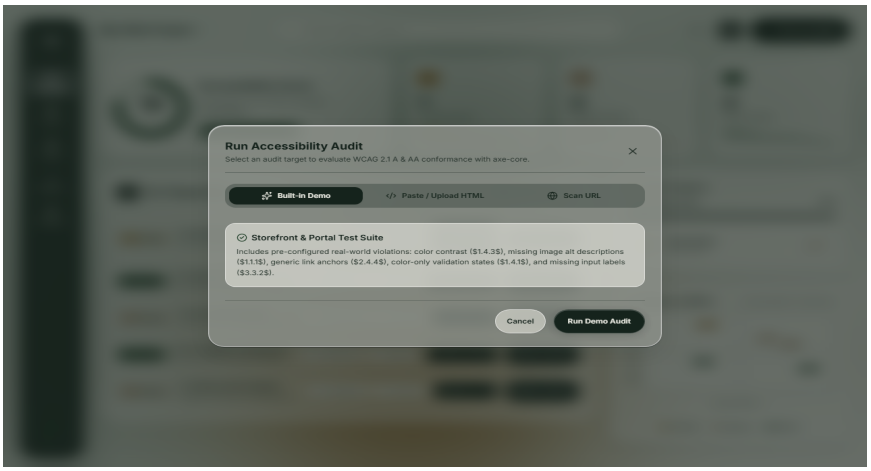


Figure A.1: Audit Execution Modal with Tabbed Sources & Serverless Proxy